



BSI Standards Publication

Determination of particle size distribution — Single particle light interaction methods

Part 4: Light scattering airborne particle counter for clean spaces

National foreword

This British Standard is the UK implementation of ISO 21501-4:2018. It supersedes BS ISO 21501-4:2007, which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee LBI/37, Particle characterization including sieving.

A list of organizations represented on this committee can be obtained on request to its secretary.

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Determination of particle size distribution — Single particle light interaction methods —

Part 4: Light scattering airborne particle counter for clean spaces

*Détermination de la distribution granulométrique — Méthodes
d'interaction lumineuse de particules uniques —*

*Partie 4: Compteur de particules en suspension dans l'air en lumière
dispersée pour espaces propres*



Reference number
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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 24, *Particle characterization including sieving*, Subcommittee SC 4, *Particle characterization*.

This second edition cancels and replaces the first edition ([ISO 21501-4:2007](http://www.iso.org/iso/21501-4:2007)), which has been technically revised.

The main changes from the previous edition are:

- [Clause 4](#) for “Principle” and [Clause 5](#) for “Basic configuration” have been added;
- “size calibration” and “verification of size setting” have been combined as “size setting error” in the requirements clause;
- “Test report” (3.11 in the previous edition) has been changed to [6.10](#) on “Reporting of test and calibration results”;
- information about uncertainties has been enriched and is now the subject of [Annex E](#).

A list of all parts in the ISO 21501 series can be found on the ISO website.

Introduction

Monitoring particle contamination levels is required in various fields, e.g. in the electronic industry, in the pharmaceutical industry, in the manufacturing of precision machines and in medical operations. Particle counters are useful instruments for monitoring particle contamination in air. The purpose of this document is to provide a calibration procedure and verification method for particle counters, so as to minimize the inaccuracy in the measurement result by a counter, as well as the differences in the results measured by different instruments.

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Determination of particle size distribution — Single particle light interaction methods —

Part 4: Light scattering airborne particle counter for clean spaces

1 Scope

This document describes a calibration and verification method for a light scattering airborne particle counter (LSAPC), which is used to measure the size distribution and particle number concentration of particles suspended in air. The light scattering method described in this document is based on single particle measurements. The typical size range of particles measured by this method is between 0,1 µm and 10 µm in particle size.

Instruments that conform to this document are used for the classification of air cleanliness in cleanrooms and associated controlled environments in accordance with [ISO 14644-1](#) and [ISO 14644-2](#), as well as the measurement of number and size distribution of particles in various environments.

The following parameters are within the scope of this document:

- size setting error;
- counting efficiency;
- size resolution;
- false count;
- maximum particle number concentration;
- sampling flow rate error;
- sampling time error;
- response rate;
- calibration interval;
- reporting results from test and calibration.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purpose of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

3.1

calibration particle

monodisperse spherical particle with a certified mean particle size, e.g. a polystyrene latex (PSL) particle, where the certified size is traceable to the International System of Units (SI), a relative standard uncertainty equal to or less than 2,5 %, and a refractive index that is approximately 1,59 at the wavelength of 589 nm (sodium D line)

3.2

counting efficiency

ratio of the number concentration measured by a *light scattering airborne particle counter (LSAPC)* (3.4) to that measured by a reference instrument for the same test aerosol

3.3

false count

apparent count per unit volume when a sample air containing no measurable particles is measured by the *light scattering airborne particle counter (LSAPC)* (3.4)

3.4

LSAPC

light scattering airborne particle counter

instrument that measures airborne particle numbers by counting the pulses as the particles pass through the sensing volume, and also particle size by scattered light intensity

Note 1 to entry: The optical particle size measured by the LSAPC is the light scattering equivalent particle size and not the geometrical size.

3.5

PHA

pulse height analyser

instrument that analyses the distribution of pulse heights

3.6

size resolution

measure of the ability of an instrument to distinguish between particles of different sizes

3.7

coincidence loss

reduction of particle count caused by multiple particles passing simultaneously through the sensing volume and/or by the finite processing time of the electronic system

3.8

test aerosol

aerosol to be used for calibration or testing of a *light scattering airborne particle counter (LSAPC)* (3.4) that is composed of *calibration particles* (3.1) suspended in clean air

3.9

MPE

maximum permissible error

limit of error

extreme value of measurement error, with respect to a known reference quantity value, permitted by specifications or regulations for a given measurement, measuring instrument, or measuring system

Note 1 to entry: This document uses decimal numbers for the requirements to MPEs to avoid confusion that may arise when relative uncertainties of test results are reported in percent figures.

4 Principle

The measurement principle of the LSAPC is based on detection of light scattered by a particle when the particle passes through an incident light beam.

The particle size is determined from the intensity of the scattered light, and the number of particles from the number of light pulses scattered by individual particles.

To be more specific, sample air is drawn from the inlet of the LSAPC at a constant flow rate, and introduced to the sensing volume of the LSAPC where a light beam is irradiated. When a particle suspended in the sample air passes through the light beam, it scatters the light, emitting a light pulse. The light pulse is detected by a photo detector, and converted to an electrical pulse. The electrical pulse height is proportional to the scattered light intensity, and depends on the optical system design, the electronic components used, and the light source. The intensity of the scattered light is dependent on the size, refractive index, and shape of the particle. If the particle is spherical, the scattered light intensity is described by the Mie theory. In order to establish a relationship between the electrical pulse height and the particle size, calibration of each LSAPC with use of particles having a well-defined size, refractive index, and shape is required.

5 Basic configuration

An LSAPC is composed typically of a light source, a sample air suction system, a sensing volume, a photoelectric conversion device, a pulse height analyser, and a display (see Figure 1). Some LSAPCs do not contain a sample air suction system and/or a display.

To make the particle size calibration possible, the LSAPC should be constructed so that pulse height distributions for calibration particles can be measured.

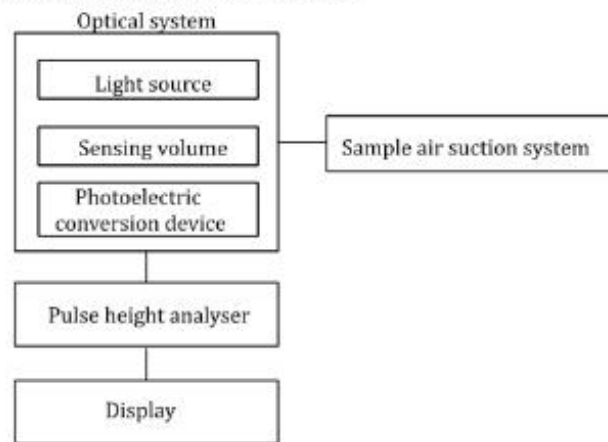


Figure 1 — Example of basic configuration of LSAPC

6 Requirements

6.1 Size setting error

The MPE for size setting in the minimum detectable particle size and other sizes specified by the manufacturer of an LSAPC is 0,10 (corresponding to 10 % of the specified size).

Size setting shall be conducted before the LSAPC is shipped from the manufacturer and when the size setting error is found not fulfilled in a periodic calibration.

A recommended procedure for size setting is described in 7.1.2. If other methods are used, their uncertainty shall be evaluated and described.

6.2 Counting efficiency

The counting efficiency shall be within 0,30 to 0,70 [corresponding to $(50 \pm 20) \%$] for calibration particles with a size close to the minimum detectable particle size, and it shall be within 0,90 to 1,10 [$(100 \pm 10) \%$] for calibration particles with a size 1,5 to 2 times larger than the minimum detectable particle size.

When calibration particles with exactly the same size as the minimum detectable particle size are not available, particles whose size is within $\pm 5 \%$ of the minimum detectable particle size may be used and the diameter of the calibration particles shall be reported.

6.3 Size resolution

The size resolution shall be less than or equal to 0,15 (corresponding to 15 % of the specified particle size), when it is evaluated using calibration particles of a certified average size specified by the manufacturer.

A recommended procedure is described in 7.3. If other methods are used, their uncertainty shall be evaluated and described.

6.4 False count

The false count per volume in cubic meters and its 95 % upper confidence limit (UCL) shall be determined according to 7.4. The 95 % UCL shall be less than or equal to the value specified and reported by the manufacturer of the LSAPC.

6.5 Maximum particle number concentration

The maximum measurable particle number concentration shall be specified by the manufacturer. The coincidence loss at the maximum particle number concentration of an LSAPC shall be less than or equal to 0,1 (10 %).

NOTE The probability of occurrence of coincidence loss increases with increasing particle number concentration.

6.6 Sampling flow rate error

The MPE of the volumetric sampling flow rate determined according to 7.6 compared to the flow rate specified by the manufacturer shall be 0,05 (corresponding to 5 %) of the specified flow rate.

6.7 Sampling time error

The MPE in the duration of the sampling time shall be 0,01 (corresponding to 1 %) of the preset value.

If the LSAPC does not have a sampling time control system, this subclause does not apply.

6.8 Response rate

The response rate of the LSAPC obtained according to the test method given in 7.8 shall be equal to or less than 0,005 (corresponding to 0,5 %).

6.9 Calibration interval

The calibration of the LSAPC should be conducted at an interval equal to or shorter than one year. The requirements should be met during the calibration interval.

6.10 Reporting of test and calibration results

The following is the minimum information that shall be recorded in a test report:

- a) date of test/calibration;
- b) test/calibration particles used;
- c) results for the parameters:
 - 1) size setting error;
 - 2) counting efficiency;
 - 3) sampling flow rate error;
 - 4) size resolution (with the particle size used);
 - 5) false count;
- d) threshold voltage values or channels of the built-in PHA corresponding to the size settings;
- e) a statement of the test/calibration method used (e.g. ISO 21501-4);
- f) report/certificate identification, test/calibration location, title and identification of test/calibration provider including signature and date;
- g) identification of customer and device under test, including how output was obtained for counting efficiency (e.g. analogue, display or digital output).

A calibration certificate shall furthermore include:

- h) identification and - if possible - statement of metrological traceability of all reference equipment and calibration particles used;
- i) relevant environmental conditions (e.g. temperature, air pressure and humidity) under which the calibration was performed;
- j) a stated uncertainty for each result for the parameters 1 to 4 with reference to the calculation method (e.g. ISO/IEC Guide 98-3) - [Annex E](#) contains procedures for evaluating the uncertainty of the results of the performance tests recommended in this document for parameters 1 and 2;
- k) a stated false count at a 95 % confidence limit (see [Annex C](#)).

NOTE Calibration certificates issued by [ISO/IEC 17025](#) accredited laboratories and covering all results for the parameters 1 to 5 are considered to comply with the requirements above.

7 Test and calibration procedures

7.1 Size setting

7.1.1 Evaluation of size setting error

Calculate the size setting error ϵ according to [Formula \(1\)](#).

$$\varepsilon = \frac{x_i' - x_i}{x_i} \quad (1)$$

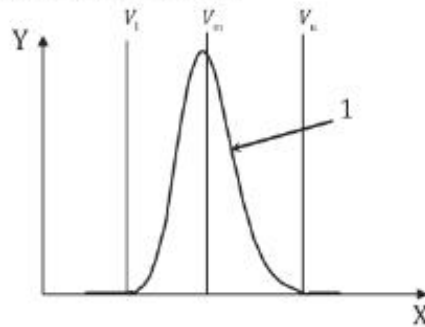
where

x_i is the size setting specified for the LSAPC;
 x_i' is the actual size setting corresponding to V_{ti} (see 7.1.2 for the meaning of V_{ti}).

7.1.2 Procedure of size setting

By use of a PHA connected to the output terminal for signal pulses of the LSAPC, or by use of a built-in PHA if one is contained as a part of the LSAPC, obtain a pulse height distribution for a test aerosol in which calibration particles are suspended. Let V_l and V_u denote the lower and upper voltage limits, respectively, of the range of pulse heights for the calibration particles (see Figure 2). The median voltage V_m of the pulse height distribution in the range from V_l to V_u shall be calculated, and is assigned to the certified size of the calibration particles, x_c .

When a built-in PHA is used, the abscissa of the pulse height distribution may be given in channel number instead of voltage. In this case, the term "voltage" above and in relevant descriptions below should be interpreted as channel number of the PHA.



Key

X	pulse height voltage
Y	frequency
1	pulse height distribution
V_l	lower voltage limit
V_m	median voltage
V_u	upper voltage limit

Figure 2 — Pulse height distribution for the test aerosol

If a noise distribution is observed in the pulse height distribution, and if it is separated distinctly from the main peak corresponding to the calibration particles, the voltages V_l and V_u shall be chosen so that the range (V_l , V_u) encompasses only the main peak [see Figure 3 a)]. If the noise distribution overlaps with the main peak, V_l and V_u shall be chosen so that the range (V_l , V_u) corresponds to the full width at half maximum of the main peak [see Figure 3 b)]. The latter way of determining V_l and V_u is allowed only when the height of the valley between the noise distribution and the main peak is at most half the main peak height.

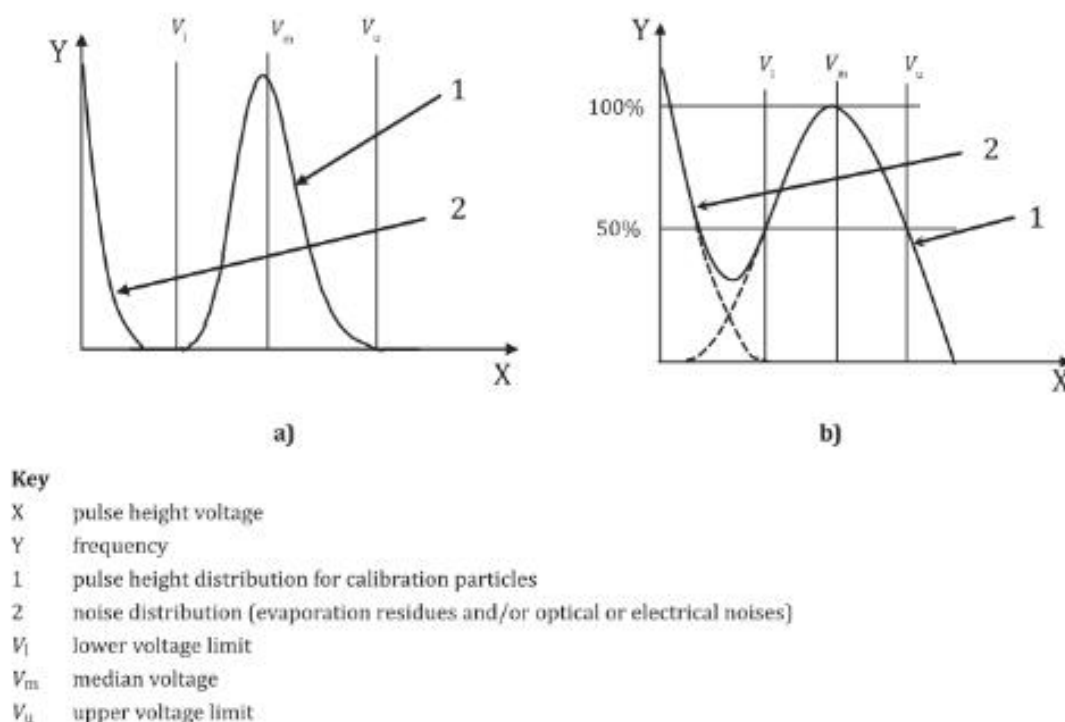


Figure 3 — Pulse height distribution for the test aerosol when noise exists

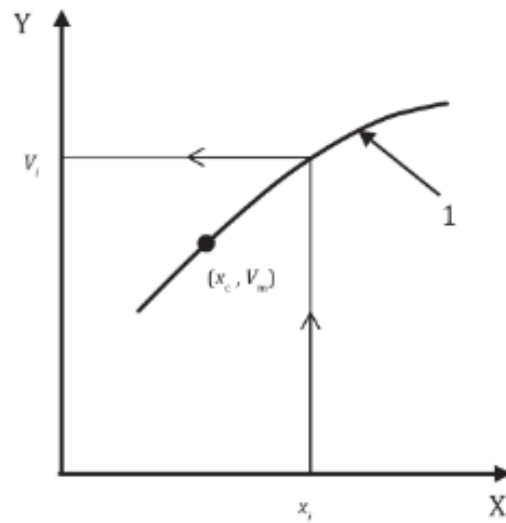
By use of the data pair (x_c, V_m) obtained in this way, or multiple data pairs (x_{cj}, V_{mj}) ($j = 1, 2, \dots$) obtained similarly for multiple calibration particles, determine the voltage values V_i ($i = 1, 2, \dots$) that correspond to the size settings (or threshold sizes) x_i given as specifications of the LSAPC (see Figure 4). In this determination, a theoretical response curve based on Mie theory may be used to calculate V_i from experimentally observed V_m .

Let V_{ti} denote the adjustable threshold voltage corresponding to x_i . For all the size settings x_i , adjust the value of V_{ti} to V_i .

NOTE 1 The response curve can be calculated according to the Mie theory when the parameter set defining the optical system of the LSAPC is available. If the parameter set of the optical system is not available, the response curve in the vicinity of x_i can still be empirically determined by fitting a simple function, e.g. a quadratic or cubic polynomial, to multiple data pairs (x_{cj}, V_{mj}) obtained for x_{cj} on either side of x_i .

NOTE 2 The detailed procedure for determining V_i can vary depending on the model of the LSAPC.

NOTE 3 V_{ij} can be the set voltage of an electric comparator used in the LSAPC, or if a built-in PHA is used, it can be the threshold channel of the built-in PHA which is intended to be assigned to x_i . For the sake of simplicity in description, it is assumed that electric comparators are employed in the LSAPC for the rest of this document, unless otherwise stated.



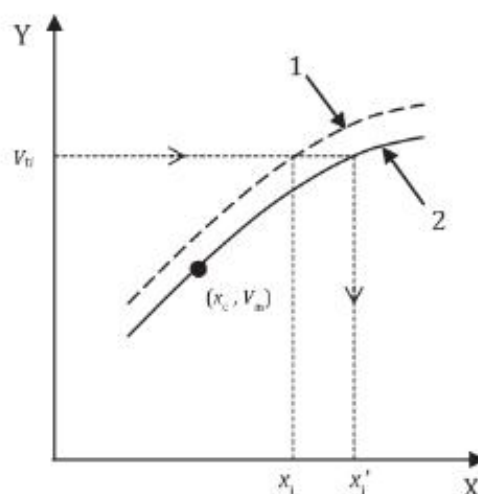
Key

X	particle size
Y	pulse height voltage
1	response curve
x_c	certified size of the calibration particles
V_m	median voltage corresponding to x_c
x_i	size setting specified for the LSAPC
V_i	voltage corresponding to x_i

Figure 4 — Size calibration

Read out the value of V_{ti} set for the electric comparator of the LSAPC. Ideally V_{ti} corresponds to x_i , but in reality V_{ti} corresponds to a particle size x_i' which may be different from x_i owing, for example, to a change of the response curve over time. Determine the actual response curve according to the procedure as described above or to another method which is scientifically documented, and determine x_i' using this curve (see [Figure 5](#)). Calculate the size setting error ϵ according to [Formula \(1\)](#) above.

NOTE 4 The expected response curve in [Figure 5](#) is a hypothetical curve on which the threshold voltages of the electric comparator, V_{ti} , would correspond exactly to the specified size thresholds x_i .



Key

X	particle size
Y	pulse height voltage
1	expected response curve
2	actual response curve
x_c	certified size of the calibration particles
V_m	median voltage corresponding to x_c
x_i	size setting specified for the LSAPC
x_i'	actual size setting corresponding to V_U
V_U	voltage read out from the electric comparator

Figure 5 — Evaluation of size setting error

7.2 Evaluation of counting efficiency

To evaluate the counting efficiency of the LSAPC, use two populations of calibration particles; one that has a size close to the minimum detectable particle size, and another that has a size 1,5 to 2 times larger than the minimum detectable particle size.

Tests with other particle sizes may be added, if it is requested by a user of the LSAPC.

Use either a condensation particle counter (CPC) combined with a differential electrical mobility classifier (DEMC) or a calibrated LSAPC as a reference instrument. The counting efficiency of the reference instrument shall have a metrological traceability to a national or international standard, or the International System of Units (SI).

NOTE 1 The condensation particle counter is also referred to as a condensation nucleus counter (CNC).

Measure the number concentrations of test aerosols suspending each of the two kinds of calibration particles with the LSAPC under test and with the reference instrument (see Annex A). Determine the counting efficiency according to Formula (2):

$$\eta = \frac{C_1}{C_0} \quad (2)$$

where

- η is the counting efficiency;
- C_0 is the particle number concentration measured by reference particle counter;
- C_1 is the particle number concentration measured by particle counter under test.

For these measurements, the particle number concentration of the test sample should be equal to or less than 25 % of the maximum particle number concentration of both the LSAPC under test and the reference instrument.

NOTE 2 When the particle concentration measured by an LSAPC is, as usually is the case, not corrected for the coincidence loss, the counting efficiency of the LSAPC depends on the particle number concentration stemming from the coincidence loss. If the maximum particle number concentration is so determined that the coincidence loss at this concentration is 0,1 (10 %) (see 6.5), and the counting efficiency η is evaluated at 0,25 (25 %) of this concentration, then the obtained value of η is smaller than the value that would be obtained in the limit of zero concentration by approximately 0,026 (2,6 %).

7.3 Evaluation of size resolution

Calculate the size resolution of the LSAPC, R , by [Formula \(3\)](#) (see also [Annex B](#)).

$$R = \frac{\sqrt{\sigma^2 - \sigma_c^2}}{x_c} \quad (3)$$

where

- R is the size resolution;
- σ is the apparent standard deviation of the size distribution of the calibration particles observed by the LSAPC;
- σ_c is the standard deviation of the size distribution of the calibration particles provided by the manufacturer of the calibration particles;
- x_c is the certified average size of the calibration particles.

NOTE Due to the uncertainties in determining σ and σ_c , σ^2 can, in some cases, be smaller than σ_c^2 . In such cases, the value of R is regarded as 0.

The particle size recommended by the manufacturer of the LSAPC should be used for this test. The standard deviation of the calibration particles, σ_c , should be known. It is recommended to determine the median voltage (or channel) V_m of the pulse height distribution for the calibration particles, as shown in [Figure 6](#), in accordance with the method given in [7.1.2](#).

Determine the lower and upper voltage limits, V_l and V_u , which correspond to 61 % of the peak height in the pulse height distribution. Using the calibration curve, determine the particle sizes x_l and x_u corresponding respectively to V_l and V_u . Calculate the absolute value of the differences, $|x_l - x_c|$ and $|x_u - x_c|$, where x_c is the certified size of the calibration particles. Let the apparent standard deviation, σ , be equal to the larger one of $|x_l - x_c|$ and $|x_u - x_c|$.

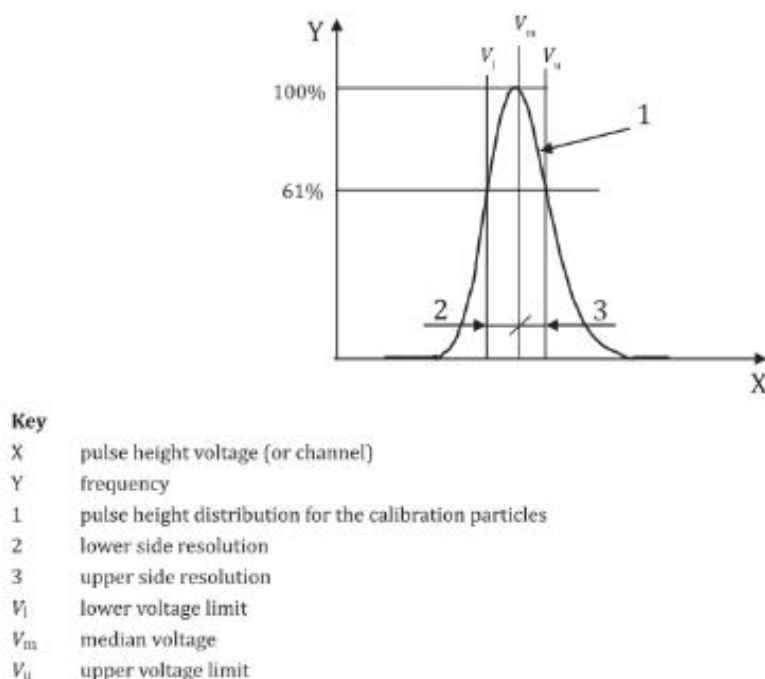


Figure 6 — Verification of size resolution

7.4 Evaluation of false count

Obtain the particle count at the size channel corresponding to the minimum detectable particle size for a certain volume of air using ambient air with filter on the inlet of the LSAPC under test. Calculate the 95 % UCL of the false count according to the procedure given in [Annex C](#). Determine the false count and its 95 % UCL by dividing them by the volume of the sample air.

7.5 Estimation of coincidence loss at the maximum particle number concentration

The coincidence loss is determined by the flow rate, the time required for particles to pass through the sensing volume and the electrical signal processing time. These values are determined by the design of the LSAPC. Coincidence loss is calculated according to [Formula \(4\)](#).

$$L = 1 - \exp(-q \cdot t_{\text{total}} \cdot C_{\text{max}}) \quad (4)$$

where

- | | |
|--------------------|--|
| L | is the coincidence loss at the maximum particle number concentration; |
| q | is the flow rate; |
| t_{total} | is the sum of the time for a particle to pass through the sensing volume and the electrical processing time; |
| C_{max} | is the maximum particle number concentration. |

7.6 Evaluation of sampling flow rate error

The sampling flow rate should be measured using either a soap bubble film flow meter, a wet gas meter, or some other type of flow meter that has a low pressure drop. The flow meter used shall have a valid

calibration certificate. The flow rate to be measured is the volumetric flow rate. Calculate the error in the sampling flow rate, ε_q , by [Formula \(5\)](#).

$$\varepsilon_q = \frac{q_m - q_s}{q_s} \quad (5)$$

where

- ε_q is the sampling flow rate error;
- q_s is the sampling flow rate specified by the manufacturer;
- q_m is the measured sampling flow rate.

When using a mass flow meter, the flow rate should be converted to a volumetric flow rate at actual conditions, taking the temperature and air pressure into account.

7.7 Evaluation of sampling time error

Sampling time is the time during which the LSAPC measures a sample (from the beginning of counting to the end of counting). Calculate the error in the sampling time, ε_t , by [Formula \(6\)](#).

$$\varepsilon_t = \frac{t_m - t_0}{t_0} \quad (6)$$

where

- ε_t is the sampling time error;
- t_0 is the sampling time preset to the LSAPC;
- t_m is the measured sampling time.

Calibrated instruments should be used for sampling time measurement.

7.8 Evaluation of response rate

Introduce into the LSAPC a test aerosol suspending calibration particles the size of which is 1,5 times larger than the minimum detectable particle size. The particle number concentration of the test aerosol should be approximately equal to the maximum particle number concentration. Set the size channel for particle counting at the minimum detectable particle size, set the LSAPC to accumulated counting and operate the LSAPC for 10 min.

After the 10 min operation, measure the particle number concentration for the period of t . The duration t shall be equal to or shorter than 60 s, and the particle count obtained during this period shall be greater than 1 000 (see [Annex D](#)). After that, switch the sample air from the test aerosol to particle-free clean air. Wait for 10 s after switching the sample air, and restart particle counting. Measure the particle number concentration for the period of t for the clean air. Calculate the response rate, R_r , by [Formula \(7\)](#) (see also [Annex D](#)).

$$R_r = \frac{C_{\text{after}}}{C_{\text{before}}} \quad (7)$$

where

R_r is the response rate;
 C_{before} is the particle number concentration observed before switching the sample air;
 C_{after} is the particle number concentration observed after switching the sample air.

Annex A (informative)

Counting efficiency

[Figure A.1](#) shows the test system for counting efficiency. The particle generator generates an aerosol that consists of dry monodisperse PSL particles suspended in clean air.

After aerosolisation of a PSL suspension, the aerosol typically contains traces of contaminants which can bias the measurement of the counting efficiency. Measurement errors should be minimized by:

- separating the PSL particles from surfactant particles, e.g. in several mixing/settling separation steps in ultrapure water before preparing the suspension for the aerosol generator;
- using a PSL suspension in the aerosol generator with very low concentration of contaminants in the liquid phase, e.g. traces of salt in ultrapure water, to a) achieve a low enough background of residue particles and b) avoid growth of PSL particles due to attachment of contaminants after evaporation of the suspension liquid droplet;
- optimising the concentration of PSL particles in the suspension to avoid measurement bias due to doublet PSL particles (two PSL particles were contained in a droplet);
- drying the aerosol to remove all suspension liquid from the surface of the PSL particles and to avoid condensation of suspension liquid vapour on the PSL particles.

After drying the aerosol, size classifying the PSL particles with a DEMC (compare [ISO 15900](#) and [ISO 27891](#)) can be applied if the background of residue particles needs to be further reduced. This may especially be necessary if the requirements in [Clause 7](#) (see [Figure 3](#)) cannot be fulfilled.

Since PSL aerosol generated from a suspension is electrostatically charged and since even DEMC-classified PSL particles are unipolarly charged, a bipolar diffusion charge conditioner (as known as aerosol neutralizer) further increases the accuracy of the measurement of the counting efficiency by minimizing particle losses in both the particle counter to be inspected and the reference particle counter.

After generation and conditioning, the PSL aerosol is fed to the particle counter to be inspected and the reference particle counter via a device (e.g. a distributing box, see [Figure A.1](#)) which shall be designed in such a way that the particle number concentration at the inlet of both particle counters is as close as possible. The uncertainty associated with the inhomogeneity in the particle number concentration should be evaluated according to the procedure given in [E.2](#) (1).

The counting efficiency is obtained by calculating the ratio of the particle number concentration measured by the particle counter under test and the particle number concentration measured by the reference particle counter. The particle number concentration of the sample should be less than 25 % of the maximum particle number concentration of both the reference particle counter and the particle counter under test.

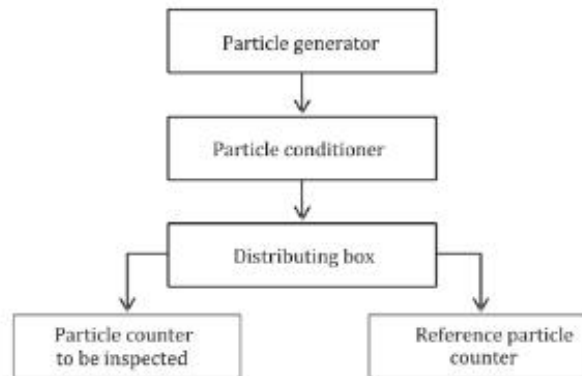


Figure A.1 — Example of counting efficiency test system

The method described above is most useful for PSL particles smaller than approximately 1 μm . If calibration with larger particles (e.g. 5 μm) is required, dry PSL particles are better suited. However, the counting efficiency may decrease considerably for particles with a diameter larger than 1 μm . The monodisperse, dry PSL powder has to be free of surfactants to avoid errors during the calibration. A dry powder disperser generates the calibration aerosol. Distributing the aerosol between the reference particle counter and the particle counter to be inspected requires special attention for larger particles since particle losses due to inertial impaction and gravitational settling become important. To minimize errors:

- use a distribution tube instead of a distribution box;
- use isokinetic and isoaxial probes to extract the calibration aerosol for both particle counters;
- use vertical tubing to connect the distribution tube with the particle counters;
- if bends in the connection tubing cannot be avoided, use a large radius of curvature (radius larger than 10 times the inner diameter of the tube);
- use metallic, grounded tubing with polished inner surface for connection;
- avoid changes in tubing diameter; in particular avoid step changes.

Annex B (informative)

Size resolution

Size resolution is defined as one standard deviation of the measured size distribution of monodisperse calibration particles expressed as the mean size of the monodisperse calibration particles.

If the distribution of calibration particles is assumed to be the Gaussian distribution,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right] \quad (\text{B.1})$$

where

$f(x)$ is the Gaussian function;
 x is the particle size;
 μ is the mean value;
 σ is the standard deviation.

When $(x - \mu) = \pm\sigma$, the ratio of density to the maximum density is $\exp\left(-\frac{1}{2}\right) = 0,61$. This is the basis for the use of 61 % in the determination of size resolution.

Annex C (informative)

False count

The probability of appearance of false counts is assumed to be described by the Poisson distribution. The Poisson distribution is defined by [Formula \(C.1\)](#):

$$P(X; \lambda) = \frac{e^{-\lambda} \lambda^X}{X!} \quad (\text{C.1})$$

where

X is the number of false counts;
 λ is the mean value of the population;
 $P(X; \lambda)$ is the probability of observing value X from a population having a mean value of λ .

The upper confidence limit, λ_U is defined by [Formula \(C.2\)](#):

$$\sum_{x=0}^X P(x; \lambda_U) = \varepsilon \quad (\text{C.2})$$

where ε is the significant level.

When the confidence limit is 95 %, ε is 0,05.

[Table C.1](#) shows the observed count and the calculated upper 95 % confidence limit. When the observed count is zero, it is possible to have up to three counts with a probability of 5 %. For example, if zero counts are observed in 15 min at the sample flow rate of 28,3 l/min, the false count rate is three counts in the volume sampled in 15 min with a 95 % confidence limit, i.e. the false count rate is 7 counts per cubic meter.

Table C.1 — Observed count and 95 % confidence limit

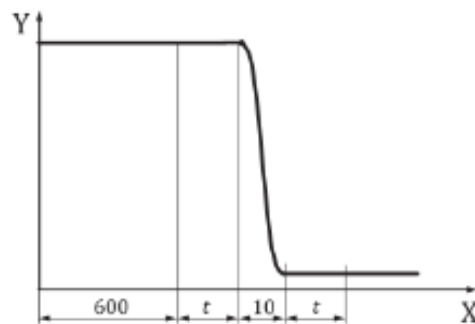
Observed count	Upper confidence limit λ_U
0	3
1	4,7
2	6,3
3	7,8
4	9,2
5	10,5
6	11,8
7	13,1
8	14,4
9	15,7
10	17,0

Annex D (informative)

Response rate

The response rate test is used to evaluate the response rate of an LSAPC when the sample air is changed to a clean air sample from a high particle number concentration sample. If the airflow in the sensor is poor, residual-particles are a cause for a poor response rate. When an LSAPC is used for a long time, there is a possibility that some particles will form sediment in the sensor.

In this test, a high particle number concentration sample is introduced to the LSAPC for 10 min. Particles are then measured for t and then the sample air is changed to clean air. After 10 s, particles are again measured for t . The ratio of the two counts for t is the response rate (see [Figure D.1](#)).



Key

- X time (seconds)
- Y particle concentration
- t length of measurement time of particles; $t \leq 60$ s

Figure D.1 — Response rate

Annex E (informative)

Procedure for evaluating the uncertainties of the results of the performance tests

E.1 Basics on measurement uncertainty

In this annex, a recommended procedure is described for evaluating the uncertainties of the results of the tests specified in 7.1 and 7.2 (Note 1). The general idea of this procedure follows ISO/IEC Guide 98-3, which is briefly summarized as follows.

Step 1) Identify the relationship between the measurand, y , and the input quantities, x_i ($i = 1, 2, \dots, N$):

$$y = f(x_1, x_2, \dots, x_N) \quad (\text{E.1})$$

This functional relationship is called the mathematical model of measurement (Notes 2 and 3).

Step 2

Evaluate the standard uncertainty $u(x_i)$ of the input quantity x_i either by Type A or Type B evaluation of uncertainty (Notes 4 and 5).

Step 3) Combine the standard uncertainties of all x_i 's to obtain the combined standard uncertainty of the measurement result, $u_c(y)$, according to the following 'law of propagation of uncertainty', (Notes 6 and 7)

$$u_c(y) = \sqrt{\sum_{i=1}^N \left(\frac{\partial f}{\partial x_i} u(x_i) \right)^2} \quad (\text{E.2})$$

Step 4) When necessary, the expanded uncertainty U is calculated according to the following equation:

$$U = k \times u_c(y) \quad (\text{E.3})$$

where k is the coverage factor. In this d, $k = 2$ is consistently used for simplicity (Note 8).

NOTE 1 The uncertainty components considered in this Annex are those relevant to the tests specified in the main body of this document. These components are considered to cover major factors that can affect measurements of particles in the real environment, but are not intended to cover all of them. Additional factors that are not considered in this Annex include the difference in optical properties between test particles and particles in the real environment, and the uncertainty associated with the determination of theoretical response functions.

NOTE 2 Input quantity is a quantity whose value is used to determine the result of measurement, or a quantity that can otherwise affect a measurement result.

NOTE 3 Although the quantities, Y and X_i , and their estimates, y and x_i , are represented by different symbols in GUM, the same symbols are used here, as far as there is no fear of confusion.

NOTE 4 If the estimate of a quantity x_i is obtained from $x_i = \bar{q}$, where $\bar{q}$ is the mean of a series of observations, q_k ($k = 1, 2, \dots, n$), then the standard uncertainty of x_i is evaluated as

$$u(x_i) = \frac{s}{\sqrt{n}} \quad (\text{E.4})$$

Here s is the experimental standard deviation of q_k given by $s = \sqrt{\sum_{k=1}^n (q_k - \bar{q})^2 / (n-1)}$, or some other estimate of the standard deviation based on an experiment conducted separately from the measurement of x_i . Uncertainty evaluation based on such statistical analysis of series of observations is called type A evaluation of uncertainty.

NOTE 5 Method of evaluation of uncertainty by means other than the statistical analysis of series of observations is called type B evaluation. Type B evaluation is conducted on the basis of available information such as data given in calibration certificates, instrument specifications, handbooks, and data obtained in the past.

NOTE 6 Formula E.2 applies to cases where there are no correlations between input quantities. When the correlations are not negligible, extra terms containing correlation coefficients are added to the right-hand side of Formula (E.2). In this document, only cases where correlations are negligible are considered.

NOTE 7 When the mathematical model (E.1) takes the form:

$$y = C \cdot x_1^{p_1} \cdot x_2^{p_2} \cdot \dots \cdot x_N^{p_N} \quad (\text{E.5})$$

where $C, p_1, p_2, \dots$ are constants, Formula (E.3) is equivalent to the following equation called the law of propagation of relative uncertainty,

$$\frac{u_c(y)}{|y|} = \sqrt{\sum_{i=1}^N \left[p_i \frac{u(x_i)}{x_i} \right]^2} \quad (\text{E.6})$$

NOTE 8 If the probability distribution characterized by y and $u_c(y)$ is a normal distribution, then the interval $y \pm U$ with $k = 2$ encompasses approximately 95 % of the distribution. The coverage factor k can have a different value depending on the actual degrees of freedom, see ISO/IEC Guide 98-3.

E.2 Uncertainty of counting efficiency, η

Let η_0 and $u(\eta_0)$ denote, respectively, the counting efficiency of a reference instrument and its standard uncertainty (Note 1). Repeat n times a parallel simultaneous measurement of a test aerosol by a particle counter, or device, under test (DUT) and the reference instrument (REF). The number n should be larger than five. The particle concentration of the test aerosol and/or the duration of a single measurement should be chosen so that the counts obtained by the REF and the DUT both exceed 1 000. In addition, as recommended in Annex A, the particle concentration should not exceed 25 % of the maximum particle number concentration specified for the REF and the DUT.

Let c_{0i} and c_i denote the particle number concentration obtained by the REF and the DUT, respectively, in the i -th repetition of the measurement. Here c_{0i} is defined as the measurement data before the correction by η_0 is applied (Note 2). Let $\bar{r}$ denote the average of the ratio $r_i = c_i / c_{0i}$. The mathematical model of the counting efficiency test, which corresponding to Formula (E.1), is given by:

$$\eta = \frac{\eta_0 \bar{r}}{\alpha} \quad (\text{E.7})$$

where α is a factor accounting for the spatial inhomogeneity in the particle concentration in the test chamber, and is defined as the ratio of the concentration at the sampling position of the DUT to that at the sampling position of the REF as shown in Figure E.1 (Note 3).

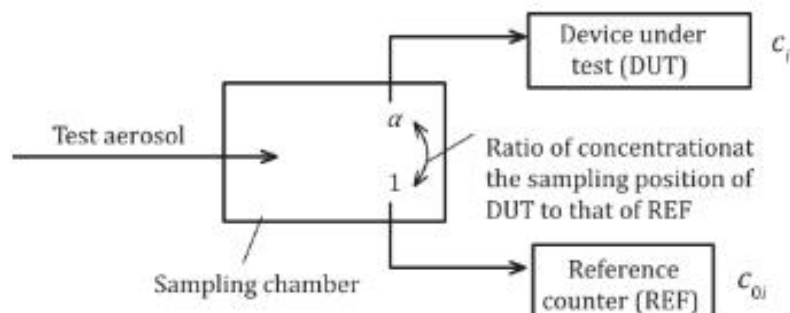


Figure E.1 — Example of experimental configuration for the uncertainty evaluation of the counting efficiency test

From [Formula \(E.6\)](#), the standard uncertainty $u_c(\eta)$ is evaluated according to [Formula \(E.8\)](#), (see Note 4).

$$\frac{u_c^2(\eta)}{\eta^2} = \frac{u^2(\alpha)}{\alpha^2} + \frac{u^2(\eta_0)}{\eta_0^2} + \frac{u^2(\bar{r})}{\bar{r}^2} \quad (\text{E.8})$$

The standard uncertainties in the right hand side of this equation, $u(\alpha)$, $u(\eta_0)$, and $u(\bar{r})$, are evaluated as follows.

(1) $u(\alpha)$

If it is known from experience or other available information that the value of α is within the interval $[1 - \Delta, 1 + \Delta]$, then evaluate $u(\alpha)$ as $u(\alpha) = \Delta / \sqrt{3}$. The value of α in the denominator in the right hand side of [Formula \(E.8\)](#) may be set approximately at 1, when the value of α is not exactly known.

If no information is available on α , it should be estimated experimentally as follows. First, prepare two particle counters, A and B. It is not necessary to know the counting efficiencies of these counters. Set the sampling positions of A and B counters at those of the REF and the DUT shown in [Figure E.1](#), respectively, and conduct a simultaneous parallel measurement by the two counters. Let c_{0A} and c_B denote the concentrations obtained by A and B, respectively. Then exchange the sampling positions of A and B, and conduct a similar measurement. Let c_{0B} and c_A denote the concentrations obtained by B and A in this configuration. The value of α is determined from the [Formula \(E.9\)](#),

$$\alpha = \sqrt{\frac{c_A c_B}{c_{0A} c_{0B}}} \quad (\text{E.9})$$

Prepare a vacuum pump, P, which has a suction capacity as large as the maximum sampling flow rate expected among possible particle counters to be tested. A particle counter instead of a pump may be used for P. Place the sampling tube inlet of P either close to the sampling position of the DUT, to that of the REF, or on some other representative locations within the test chamber, and operate the pump. The purpose of the pump is to produce various flow patterns that may arise in actual counting efficiency tests. Evaluate α in the manner described above, with the sampling tube inlet of P at least on five different positions in the chamber. Let Δ be the maximum among the obtained values of $|\alpha - 1|$.

Determine $u(\alpha)$ as $u(\alpha) = \Delta / \sqrt{3}$.

(2) $u(\eta_0)$

The standard uncertainty $u(\eta_0)$ is obtained from the calibration certificate of the REF.

The standard uncertainty $u(\eta_0)$ is obtained from the calibration certificate of the REF.

(3) $u(\bar{r})$

Calculate the experimental standard deviation of r_i by

$$s_r = \sqrt{\sum_{i=1}^n (r_i - \bar{r})^2 / (n-1)} \quad (\text{E.10})$$

Determine $u(\bar{r})$ by

$$u(\bar{r}) = s_r / \sqrt{n} \quad (\text{E.11})$$

The standard uncertainty given in [Formula \(E.11\)](#) represents variation in the counting efficiency during short-term repeated measurements. When the variation in the counting efficiency over different measurement days is significantly larger than the variation over repeated measurements, it is desirable to conduct an experiment in which the daily variation can be evaluated; for example, conduct three repeated measurements for each of three different days. From the data obtained in this way, the variance for the day effect can be evaluated separately from the variance of the repeated measurements using the analysis of variance (ANOVA) method.

NOTE 1 Here η_0 is the value at the certified size of the test particles, x .

NOTE 2 Because C_0 given in [Formula \(2\)](#) represents the concentration obtained *after* the correction by η_0 is applied, the relation $C_0 = c_{0r} / \eta_0$ holds. The use of c_{0r} here is to take the uncertainty of η_0 into consideration in the uncertainty analysis.

NOTE 3 The best estimate of α is set at 1, but the deviation of α from 1 is taken into account as the uncertainty of α .

NOTE 4 When the DUT is used to measure particles having a very narrow size distribution around the smallest detection limit, an additional uncertainty component can be significant that is associated with the fact that the certified size of the test particles do not, in general, coincide with the size corresponding to the smallest detection limit of the DUT. This additional uncertainty, $u_x(\eta)$, is evaluated as

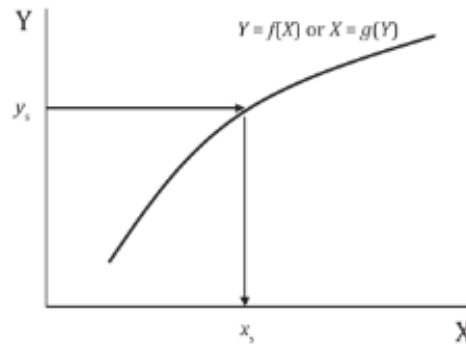
$$u_x(\eta) = \left| \frac{d\eta}{dx} \right| u(x) \quad (\text{E.12})$$

where the sensitivity coefficient $|d\eta/dx|$ is given approximately by $|d\eta/dx| = 1/(\sqrt{2\pi}\sigma)$, with σ being the standard deviation representing the width of the apparent size distribution spectrum of the test particles that would be obtained by the DUT (expressed in the unit of particle size). The uncertainty $u(x)$ in the right hand side of [Formula \(E.12\)](#) is given by $u(x) = \sqrt{(x_c - x_0)^2 + u^2(x_c)}$, where x_c is the certified size of the test particles, $u(x_c)$ is its standard uncertainty that is obtained from the calibration certificate of the test particles, and x_0 is the smallest detection size indicated in the DUT specifications.

In this document, it is assumed that the DUT is used to measure particles with relatively broad size distributions such that $u_x(\eta)$ is negligibly small compared with $u_c(\eta)$ in [Formula \(E.8\)](#), and the contribution of $u_x(\eta)$ is not taken into consideration. It is further assumed that the apparent standard deviation of the size distribution of the calibration particles observed by the DUT is large compared to the actual standard deviation of the size distribution of the calibration particles, the size setting error and the uncertainty of the size of the reference particles.

E.3 Uncertainty of the size setting error, ε

Let $Y = f(X)$ be the functional relationship of the calibration curve shown in [Figure 5](#), and $X = g(Y)$ be its inverse function (see [Figure E.2](#)).



Key

Y pulse height voltage
X particle size

Figure E.2 — Response curve

The mathematical model of measurement of ε is given by [Formula \(1\)](#). By applying [Formula \(E.2\)](#) to this formula, the combined standard uncertainty $u_c(\varepsilon)$ is determined approximately by

$$u_c^2(\varepsilon) = \frac{1}{x_r^2} \left[u^2(x_c) + \frac{u^2(y_r) + u^2(y_p)}{b^2} \right] \quad (\text{E.13})$$

where $b = \left(\partial f / \partial X \right)_{X=x_r}$. The standard uncertainties $u(x_c)$, $u(y_r)$, and $u(y_p)$ in the right hand side of [Formula \(E.13\)](#) are determined as follows.

(1) $u(x_c)$

The standard uncertainty $u(x_c)$ represents the uncertainty of the certified diameter of the calibration particles used in the evaluation of ε , and it is obtained from the calibration certificate of the calibration particles.

(2) $u(y_r)$

The standard uncertainty $u(y_r)$ represents the uncertainty of the voltage setting y_r associated with the nominal value of the size channel threshold, x_r . It consists of the uncertainty of the voltmeter scale, $u_t(y_r)$, which is obtained from the calibration certificate of the voltmeter, and the uncertainty associated with the repeatability in the experimental determination of y_r . Let n_t be the number of repetitions of measurement for experimental determination of y_r , and s_t be the experimental standard deviation of these repeated measurements (Note 1). The uncertainty $u(y_r)$ is determined from

$$u(y_r) = \sqrt{u_t^2(y_r) + \frac{s_t^2}{n_t}} \quad (\text{E.14})$$

The second term in the square root of the right-hand side of [Formula \(E.14\)](#) represents variation in y_r during short-term repeated measurements. When the variation in y_r over different measurement days is significantly larger than the variation over repeated measurements, it is desirable to conduct an experiment in which the daily variation can be evaluated.

(3) $u(y_p)$

The standard uncertainty $u(y_p)$ represents the uncertainty of the median voltage y_p of the pulse height distribution for the test particles obtained to determine the response curve of the DUT. It consists

of the uncertainty of the voltage scale of the pulse height analyser, $u_{PHA}(y_p)$, which is obtained, for example, from the specifications of the pulse height analyser, and the uncertainty associated with the repeatability in the experimental determination of y_p . Let n_{PHA} be the number of repetitions of measurements for experimental determination of y_p , and s_{PHA} be the experimental standard deviation of these repeated measurements. The uncertainty $u(y_p)$ is determined by

$$u(y_p) = \sqrt{u_{PHA}^2(y_p) + \frac{s_{PHA}^2}{n_{PHA}}} \quad (E.15)$$

E.4 Consideration of test uncertainties in the requirements for LSAPCs

MPE values and test criteria in [Clause 6](#) of this document are intended to be applied without taking the test uncertainty into account (see [Table E.1](#)). If taking an actual stated uncertainty from a test report/calibration certificate into account the MPE values and test criteria in [Clause 6](#) may be modified accordingly.

Table E.1 — Test criteria without consideration to uncertainties

Name of quantity	Target value	Requirement	Relevant subclause
Size setting error, ε	0	0,10 ($ \varepsilon \leq 0,10$)	6.1 7.1
Counting efficiency, η , at a size close to the minimum detectable particle size	0,50	0,20 ($0,30 \leq \eta \leq 0,70$)	6.2 7.2
Counting efficiency, η , at a size 1,5 to 2 times larger than the minimum detectable particle size	1,00	0,10 ($0,90 \leq \eta \leq 1,10$)	6.2 7.2
Size resolution, R	0	0,15 ($R \leq 0,15$)	6.3 7.3
False counts observed per m ³	As per manufacturer's specification	As per manufacturer's specification	6.4 7.4
Coincidence loss at the maximum particle number concentration, L	0	0,10 ($L \leq 0,10$)	6.5 7.5
Sampling flow rate error, ε_q	0	0,05 ($ \varepsilon_q \leq 0,05$)	6.6 7.6
Sampling time error, ε_t	0	0,01 ($ \varepsilon_t \leq 0,01$)	6.7 7.7
Response rate, R_r	0	0,005 ($R_r \leq 0,005$)	6.8 7.8

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- [6] [ISO 15900](#), *Determination of particle size distribution — Differential electrical mobility analysis for aerosol particles*
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- [10] JIS B 9921:2010, *Light scattering airborne particle counter for clean spaces*